

Teacher's Signature: NH

Date: _____

Physics Laboratory

United International University

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Date: _____

Experiment No. 02

Name of the Experiment: Determination of the value of the Acceleration due to Gravity (g) with the help of a compound (bar) pendulum.

Theory:

Compound pendulum is a rigid body of any shape free to turn about a horizontal axis. (See figure)

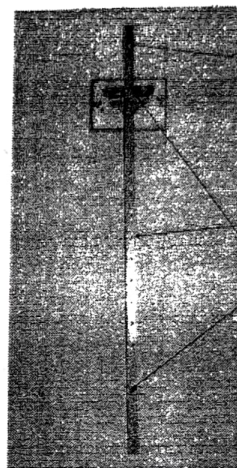
The time period (T) of a compound pendulum is given by,

$$T = \sqrt{\frac{K^2 + l^2}{gl}} \dots \dots \dots (1)$$

where,

l = the distance of the point of suspension from the center of gravity,

K = Radius of gyration of the pendulum about an axis passing through the Center of Gravity (C.G.)



• Bar pendulum

A uniform rectangular metallic bar (~1m long), with holes drilled along its length (~5 cm apart)

CG in the middle of the bar

2 knife edges symmetrically placed on either side of CG to suspend it at various distances from CG

Since the periodic time of a simple pendulum is given by $T = \sqrt{\frac{L}{g}}$, the period of the rigid body (compound pendulum) is the same as that of a simple pendulum of length,

$$L = \frac{l^2 + K^2}{l} = l + \frac{K^2}{l} \dots \dots \dots (2)$$

This length (L) is known as the length of the *Simple Equivalent Pendulum*. The expression for L can be written as a quadratic in l . Thus from equation (2),

$$l^2 - lL + K^2 = 0 \dots \dots \dots (3)$$

By solving the above equation, the two distinct roots obtained will be l_1 and l_2 for which the body has equal times of vibration. From the theory of quadratic equations,

$$l_1 + l_2 = L \text{ and } l_1 l_2 = K^2$$

As the sum and the products of two roots are positive, the two roots are both positive. This means that there are two positions of the center of suspension on the same side of C.G. about which the periods (T) would be the same. On the other side of the C.G., similarly there will be two more points of suspension, about which the time periods (T) will again be the same. Thus, there are altogether *four* points, *two* on either side of the C.G., about which the time periods of the pendulum are the same (T). The distance between two such points, asymmetrically situated on either side of the C.G., will be the length (L) of the simple equivalent pendulum. If the period of oscillation about these points are T , then

from the expression, $T = \sqrt{\frac{L}{g}}$, we get,

$$g = 4\pi^2 \frac{L}{T^2} \dots \dots \dots (4)$$

By finding L graphically, and determining the value of the period T , the acceleration due to gravity (g) at the place of the experiment, can be measured.

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Apparatus:

- Bar pendulum
- Meter scale
- Stop watch

Experimental Data:

(A) Determination of the Period (T) and Distance (d) of the knife-edge from one fixed end

At the top	Hole no.	Distance of knife-edge (d) from the fixed end (cm.)	Time of 10 oscillations (sec.)	Mean time taken (sec.)	Time period T (sec.)
On one side of C.G.	1	10	19.59	19.595	1.9595
			19.60		
	2	15	17.07	17.125	1.7125
			17.18		
	3	20	16.05	16.05	1.605
			16.05		
	4	25	15.59	15.635	1.5635
			15.68		
	5	30	15.62	15.63	1.563
			15.64		
	6	35	15.31	15.37	1.537
			15.43		
	7	40	15.69	15.77	1.577
			15.85		
	8	45	16.15	16.175	1.6175
			16.20		
On other side of C.G.	1	10	19.11	19.125	1.9125
			19.14		
	2	15	16.81	16.885	1.6885
			16.96		
	3	20	15.77	15.815	1.5815
			15.86		
	4	25	15.37	15.42	1.542
			15.47		
	5	30	15.32	15.365	1.5365
			15.41		
	6	35	15.55	15.55	1.555
			15.55		
	7	40	15.85	15.935	1.5935
			16.02		
	8	45	16.16	16.195	1.6195
			16.23		

(B) Determination of the value of 'g' from graph

No. of Obs.	Length AC (cm.)	Length BD (cm.)	Length of Eq. Simple Pendulum, $L = \frac{AC + BD}{2}$ (cm.)	Corresponding value of time period (T) from graph (sec.)	$g = 4\pi^2 \frac{L}{T^2}$ (cm./sec. ²)	Mean value of g (cm./sec. ²)
1	61	61.5	61.25	1.57	980.99	976.58
2	62.5	61.5	62.0	1.59	993.55	
3	64	63	63.5	1.62	955.22	

Calculation:

$$g_1 = 4\pi^2 \frac{L}{T^2} = 4\pi^2 \frac{61.25}{1.57^2} = 980.99 \text{ cm/s}^2$$

$$g_2 = 4\pi^2 \frac{L}{T^2} = 4\pi^2 \frac{62.0}{1.59^2} = 993.55 \text{ cm/s}^2$$

$$g_3 = 4\pi^2 \frac{L}{T^2} = 4\pi^2 \frac{63.5}{1.62^2} = 955.22 \text{ cm/s}^2$$

Therefore, $g = \frac{g_1 + g_2 + g_3}{3} = 976.58$

$$\therefore \text{Difference (\%)} = \frac{\text{Experimental Result} - \text{Theoretical Result}}{\text{Theoretical Result}} \times 100\% = 0.45\%$$

$$\therefore \text{Accuracy (\%)} = 100\% - \% \text{ Difference} = 99.55\%$$

Result:

The acceleration due to gravity is, $g = 976.58 \text{ cm/s}^2$

Discussions:

Q: What is acceleration due to gravity? What is the physical significance of acceleration due to gravity?

→ Acceleration due to gravity g is the acceleration an object experiences when it moves freely only under the influence of gravitational force of a massive body $g = \frac{GM}{r^2}$

→ • g is the gravitational field intensity $g = \frac{F}{m}$

• Determines the motion of freely falling bodies.

• Determines the weight of a body $w = mg$

• Determines orbital and escape velocities.

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Q: What are center of suspension and center of oscillation of a compound pendulum? What is simple equivalent length?

- The center of suspension of a compound pendulum is the point about which the pendulum is actually hang the allowed to oscillate. It is the fixed pivot point and the moment of inertia of the pendulum is taken about this point.
- The center of oscillation is a point on the pendulum where if the pendulum were suspended from that point, it would have the same time period.
- A compound pendulum behaves like a simple pendulum of an effective length known as the simple equivalent length $L = \frac{I}{m\bar{r}}$, $T = 2\pi\sqrt{\frac{L}{g}}$

Q: If a body is released from the roof top of UIU which is 25m above from the earth surface, calculate the time required for the body to touch the ground with g you have found in this experiment.

experiment $g = 9.76 \text{ m/s}^2$

$S = 25 \text{ m}$; $t = ?$

So, $S = \frac{1}{2}gt^2$

$\Rightarrow 25 \times 2 = 9.76 \times t^2$

$\Rightarrow t^2 = \frac{50}{9.76} = 5.122 \quad \therefore t = 2.26 \text{ sec.}$

Q: What are the advantages of Compound Pendulum over Simple Pendulum?

1. More accurate time period.
2. Easier to setup experimentally.
3. Less affected by Air resistance.
4. Use for precise determination of g.
5. More stable and convenient.



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